Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/26/2007
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Project Title:	Supported Capillary Pipes For Countercurrent Two-Phase Contacting		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	99164-2814	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	11/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Marston, Philip (Washington State University)		
Grant/Contract No.:	NNC04GA65G		
Performance Goal No.:			
Performance Goal Text:			

The goal of this research is to examine the stability and dynamics of open liquid channels with large amounts of free surface area that can sustain open-channel capillary-driven flow for applications to chemical processes necessary for life support on long-duration manned space missions. The free surface is stabilized by a support structure and thus we call the liquid channel a supported capillary pipe (SCP). Support structures considered include polygonal wire arrays and helices although primary emphasis is on helically supported capillary systems. Chemical processing equipment could be useful on long-duration manned space missions for water recycling, air purification and possibly for other purposes. The chemical processing equipment must be compact, light weight and highly reliable. Many processes used on earth for chemical separation or purification involve contacting a liquid with a gas in a piece of equipment that creates large amounts of surface area between the two fluids. These traditional processes depend on gravity for proper operation. Furthermore, process equipment used in a chemical plant on earth is not necessarily optimized for compactness or

Task Description:	minimal weight. A new idea for liquid/gas contacting was proposed by the PI which would allow for efficient operation even in the absence of gravity and is expected to be compact and light weight. How well this new process works depends on the ability to flow liquid slowly through support structures of minimal weight per volume that allow for large free surface area per volume. The liquid clings to the support structure and the free surface acts as the wall of a pipe. The surface area created between liquid and gas is just what is needed for the mass transport that is crucial for purification processes. It is proposed that by applying a pressure difference between the ends of this so-called supported capillary pipe, the fluid can be made to flow at sufficient rates for chemical processing purposes. The basic physics behind this is the same as that behind the phenomenon of wicking in a fibrous material in which a liquid can travel upward against gravity as happens in an oil lamp or candle. The primary difference is that the structures proposed here can be larger in scale since they will operate in reduced gravity. Also, even when conditions are such that the liquid is not naturally pulled into the structures as happens in wicking, it is possible to push the liquid through the structures with an enhanced entrance pressure. The structures to be examined consist of polygonal arrays of wires or helical structures. Experimentation by Don Petit on the International Space Station (Expedition 6) proved that a loop of thin wire could very easily support large volumes of water with a large amount of surface area. The loop was greater than 2 inches in diameter. The supported water was very stable to large amplitude shaking and even to boiling of the water by insertion of a soldering iron into the supported lens of liquid. The wire structures to be investigated in this work should create considerably more stable capillary structures than the large single wire loop investigated by Petit. The main objectives outlined in the proposal were as follows: 1. Determine the stability characteristics of supported capillary pipes under static conditions. 2. Study the flow characteristics in capillary pipes including initiation, steady-state flow and the effect of pressure transients. 3. Study capillary oscillations on a supported capillary pipe. 4. Explore the ability to stabilize capillary pipes using dielectrophoretic forces.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The structures studied here might be useful for microfluidic devices which are currently of significant interest for portable medical devices and other applications on earth. They may also be useful for microchannel chemical processing which is a way to intensify chemical processing or for intensified heat transfer from computer chips.
Task Progress:	The goal of this research was to examine the stability and dynamics of open liquid channels with large amounts of free surface area that could sustain open-channel capillary-driven flow for applications to chemical processes necessary for life support on long-duration manned space missions. The free surface is stabilized by a support structure and thus we call the liquid channel a supported capillary pipe (SCP). Support structures that were considered included polygonal wire arrays and helices although primary emphasis was on helically supported capillary systems. A detailed theory of the static stability of liquid channels supported by helical wires has been worked out. The theory is supported by comparison with careful experimental measurements of the stability envelope in simulated zero gravity. The measured range of stability was consistently less than predicted and this is understood in terms of nonuniform wetting of the support structure along its length. A very important consideration for practical application of SCPs has to do with how easily they can be established by pumping liquid into the wire support structures. The kinetics of this flow is expected to be governed by the Washburn equation, which applies generally to capillary driven flows. A theory for the driving force of the penetrating flow has been developed and an upper limit on the helix pitch for which injection from one end of the structure is possible is predicted. This predicted critical pitch has been experimentally confirmed. Continuous liquid channels can be established beyond the injection limit by injecting at a smaller pitch and then lengthening the structure but in this case a local instability leads to a catastrophic propagating instability that largely empties the structure. Experimental observation of penetration dynamics is consistent with the predictions of the Washburn equation. A demonstration experiment has been performed to show that a steady flow of 1 molar NaOH through a horizontal 1/8 inch diameter spring can be used to absorb CO2 gas at a significant rate. This is strong evidence that these structures can be used for chemical processing in space. The static stability results are expected to place limits on achievable flow rates, however, the flow rates that have already been demonstrated in small horizontal springs are comparable in terms of superficial velocity to typical superficial velocities in industrial gas-liquid contacting equipment.
Bibliography Type:	Description: (Last Updated: 09/14/2018)
Articles in Peer-reviewed Journals	Lowry BJ, Thiessen DB. "Fixed contact line helical interfaces in zero gravity." Phys. Fluids, 2007 Feb(2);19:022102. http://dx.doi.org/10.1063/1.2710518 , Feb-2007
Dissertations and Theses	Bernate J. "Static Stability of Helically Supported Fluid Interfaces at Zero Bond Number." Master's Thesis, Washington State University, August 2006. , Aug-2006
Papers from Meeting Proceedings	Bernate J, Thiessen DB. "Open-Channel Capillary Flow in Minimal Support Structures." Presented at AIChE 2006 Annual Meeting, San Francisco, CA, Nov. 12-17, 2006. AIChE 2006 Annual Meeting, November 2006, paper 316l. , Nov-2006